

Word Problems & Reasoning

ANSWER KEY



Number and digit reasoning

- 1 A number decreased by 9 equals 34. Find the number.
 $x - 9 = 34 \Rightarrow x = 43$.
 - 2 Three times a number, minus 7, equals 32. Find the number.
 $3x - 7 = 32 \Rightarrow 3x = 39 \Rightarrow x = 13$.
 - 3 The sum of two numbers is 58. One number is 12 more than the other. Find both numbers.
Let the smaller be x : $x + (x + 12) = 58 \Rightarrow 2x = 46 \Rightarrow x = 23$. The numbers are 23 and 35.
 - 4 A two-digit number has digit sum 11. Reversing its digits gives a number 45 more than the original. Find the original number.
Let the digits be t (tens) and u (units): $t + u = 11$ and $(10u + t) - (10t + u) = 45 \Rightarrow u - t = 5$. Solving together: $u = 8, t = 3$. The original number is 38.
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Age problems

- 5 A father is 4 times as old as his daughter. In 5 years, he will be 3 times as old as she will be. Find their current ages.
Let the daughter be x , father $4x$: $4x + 5 = 3(x + 5) \Rightarrow 4x + 5 = 3x + 15 \Rightarrow x = 10$. Daughter is 10, father is 40.
 - 6 Sara is twice as old as her brother. In 6 years, the sum of their ages will be 42. Find their current ages.
Let the brother be x , Sara $2x$: $(x + 6) + (2x + 6) = 42 \Rightarrow 3x = 24 \Rightarrow x = 10$. Brother is 10, Sara is 20.
 - 7 Ahmed is 5 years older than Salem. Seven years ago, Ahmed was twice as old as Salem. Find their current ages.
Let Salem be x , Ahmed $x + 5$: $(x + 5 - 7) = 2(x - 7) \Rightarrow x - 2 = 2x - 14 \Rightarrow x = 12$. Salem is 12, Ahmed is 17.
 - 8 The sum of a mother's and daughter's ages is 50. The mother is 20 years older than the daughter. Find both ages.
Let the daughter be d : $d + (d + 20) = 50 \Rightarrow 2d = 30 \Rightarrow d = 15$. Daughter is 15, mother is 35.
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Distance, rate and time

- 9 A train travels 180 km in 2 hours. At the same speed, how far does it travel in 5 hours?
Speed = $180 \div 2 = 90$ km/h. Distance = $90 \times 5 = 450$ km.

- 10** Two cities are 350 km apart. A car leaves one city for the other at 70 km/h. How long does the trip take?
Time = $350 \div 70 = 5$ hours.
- 11** Two cyclists start at the same point and ride in opposite directions, at 15 km/h and 20 km/h. After how many hours will they be 105 km apart?
Combined speed = $15 + 20 = 35$ km/h. Time = $105 \div 35 = 3$ hours.
- 12** A plane flies against a headwind at 480 km/h and covers 1,440 km. How many hours does the flight take?
Time = $1,440 \div 480 = 3$ hours.
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Work-rate problems

- 13** Worker A can paint a room in 6 hours; Worker B can paint the same room in 3 hours. Working together, how long will it take them?
Combined rate = $\frac{1}{6} + \frac{1}{3} = \frac{1}{6} + \frac{2}{6} = \frac{3}{6} = \frac{1}{2}$ room/hour. Time = 2 hours.
- 14** A pipe fills a tank in 10 hours; a second pipe fills it in 15 hours. Working together, how long to fill the tank?
Combined rate = $\frac{1}{10} + \frac{1}{15} = \frac{3}{30} + \frac{2}{30} = \frac{5}{30} = \frac{1}{6}$. Time = 6 hours.
- 15** 6 workers can complete a job in 8 days. How many days would 4 workers take, at the same rate?
Total work = $6 \times 8 = 48$ worker-days. With 4 workers: $48 \div 4 = 12$ days.
- 16** A pipe fills a tank in 12 hours, but a leak can empty a full tank in 24 hours. If both act at once, how long to fill the tank?
Net rate = $\frac{1}{12} - \frac{1}{24} = \frac{2}{24} - \frac{1}{24} = \frac{1}{24}$. Time = 24 hours.
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Mixture and money problems

- 17** A jeweler mixes gold worth 200 riyals/gram with silver worth 50 riyals/gram to make 10 grams of a mixture worth 110 riyals/gram. How many grams of gold are used?
Let g be grams of gold: $200g + 50(10 - g) = 110 \times 10 \Rightarrow 150g = 600 \Rightarrow g = 4$ grams.
- 18** A shopkeeper has 15 riyals in 1-riyal and 2-riyal coins, 10 coins in total. How many of each coin does she have?
Let x be 1-riyal coins, y be 2-riyal coins: $x + y = 10$, $x + 2y = 15$. Subtracting: $y = 5$, $x = 5$.
- 19** A solution is 40% acid. How many liters of pure water must be added to 20 liters of this solution to dilute it to 25% acid?
Acid amount stays $0.4 \times 20 = 8$ liters. $\frac{8}{20 + w} = 0.25 \Rightarrow 20 + w = 32 \Rightarrow w = 12$ liters.

- 20** Event tickets cost 30 riyals for adults and 18 riyals for children. 150 tickets were sold for 3,780 riyals in total. How many adult tickets were sold?
Let a be adults, so $150 - a$ children: $30a + 18(150 - a) = 3,780 \Rightarrow 12a = 1,080 \Rightarrow a = 90$ adult tickets.
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Logical deduction

- 21** Ali, Basim and Khalid have 36 books together. Ali has twice as many as Basim, and Khalid has 6 more than Ali. Find how many books each has.
Let Basim = x , Ali = $2x$, Khalid = $2x + 6$: $x + 2x + 2x + 6 = 36 \Rightarrow 5x = 30 \Rightarrow x = 6$. Basim has 6, Ali has 12, Khalid has 18.
- 22** A number between 50 and 70 is divisible by both 3 and 4, but not by 8. Find the number.
Multiples of 12 between 50 and 70: only 60. Is 60 divisible by 8? $60 \div 8 = 7.5$, no. The number is 60.
- 23** In a class, every student plays football or basketball (or both). 25 play football, 18 play basketball, and 10 play both. How many students are in the class?
By inclusion-exclusion: $25 + 18 - 10 = 33$ students.
- 24** Four consecutive even integers sum to 92. Find the largest one.
Let the integers be $n, n + 2, n + 4, n + 6$: $4n + 12 = 92 \Rightarrow n = 20$. The largest is 26.
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Working backward

- 25** After spending half his money on a book, then 20 riyals on lunch, Yousef has 30 riyals left. How much did he start with?
Let the start be x : $\frac{x}{2} - 20 = 30 \Rightarrow \frac{x}{2} = 50 \Rightarrow x = 100$ riyals.
- 26** A number is tripled, then 8 is subtracted, giving 25. Find the original number.
 $3x - 8 = 25 \Rightarrow 3x = 33 \Rightarrow x = 11$.
- 27** After a 15% pay cut, an employee's salary is 3,400 riyals. Find the salary before the cut.
Original $\times 0.85 = 3,400 \Rightarrow$ original = $3,400 \div 0.85 = 4,000$ riyals.
- 28** A water tank is $\frac{1}{4}$ full. After adding 60 liters, it becomes $\frac{5}{8}$ full. Find the tank's total capacity.
 $60 = C\left(\frac{5}{8} - \frac{1}{4}\right) = C\left(\frac{3}{8}\right) \Rightarrow C = 60 \times \frac{8}{3} = 160$ liters.
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Mixed word problems

- 29** A rectangular field is twice as long as it is wide, and its perimeter is 90 m. Find its length and width.
Let width = w , length = $2w$: $2(w + 2w) = 90 \Rightarrow 6w = 90 \Rightarrow w = 15$ m. Length = 30 m.

- 30** Layla buys 3 notebooks and 2 pens for 29 riyals. Fahad buys 1 notebook and 4 pens for 23 riyals, at the same prices. Find the price of one notebook and one pen.
Let n be the notebook price, p the pen price: $3n + 2p = 29$, $n + 4p = 23 \Rightarrow n = 23 - 4p$. Substituting:
 $3(23 - 4p) + 2p = 29 \Rightarrow -10p = -40 \Rightarrow p = 4$, $n = 7$. Notebook: 7 riyals, pen: 4 riyals.
- 31** A car rental costs 80 riyals per day plus 0.50 riyals per km driven. A 3-day rental cost 435 riyals in total. How many km were driven?
 $80 \times 3 + 0.5k = 435 \Rightarrow 240 + 0.5k = 435 \Rightarrow 0.5k = 195 \Rightarrow k = 390$ km.
- 32** The sum of three consecutive odd integers is 63. Find the integers.
Let the smallest be n : $n + (n + 2) + (n + 4) = 63 \Rightarrow 3n + 6 = 63 \Rightarrow n = 19$. The integers are 19, 21, 23.
- 33** A shop sells notebooks for 6 riyals each, with a "buy 3, get 1 free" offer. Layla needs 20 notebooks. How much will she pay?
Every group of 4 notebooks costs for 3: $20 \div 4 = 5$ groups, each costing $3 \times 6 = 18$ riyals. Total
 $= 5 \times 18 = 90$ riyals.
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More age and number problems

- 34** The sum of a number and its square is 42. Find the positive value of the number.
 $x + x^2 = 42 \Rightarrow x^2 + x - 42 = 0 \Rightarrow (x + 7)(x - 6) = 0$. The positive value is $x = 6$.
- 35** A number's triple exceeds its double by 15. Find the number.
 $3x = 2x + 15 \Rightarrow x = 15$.
- 36** In 10 years, Noura will be 3 times as old as she was 2 years ago. Find her current age.
Let her current age be x : $x + 10 = 3(x - 2) \Rightarrow x + 10 = 3x - 6 \Rightarrow 16 = 2x \Rightarrow x = 8$.
- 37** The sum of two consecutive multiples of 5 is 75. Find the larger one.
Let the numbers be $5k$ and $5k + 5$: $10k + 5 = 75 \Rightarrow k = 7$. The numbers are 35 and 40; the larger is 40.
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More rate and proportion problems

- 38** A photocopier prints 45 pages per minute. How many minutes does it take to print 810 pages?
 $810 \div 45 = 18$ minutes.
- 39** A recipe for 6 people uses 450 g of rice. How much rice is needed for 15 people, at the same rate per person?
Rice per person $= 450 \div 6 = 75$ g. For 15 people: $15 \times 75 = 1,125$ g.

- 40** A factory produces 1,200 items in 8 hours. At the same rate, how many items would it produce in a 10-hour shift?

Rate = $1,200 \div 8 = 150$ items/hour. In 10 hours: $10 \times 150 = 1,500$ items.

- 41** Three taps together fill a pool in 4 hours. Two of the taps alone would take 6 hours. How long would the third tap alone take to fill the pool?

Combined rate = $\frac{1}{4}$. Two-tap rate = $\frac{1}{6}$. Third tap rate = $\frac{1}{4} - \frac{1}{6} = \frac{3}{12} - \frac{2}{12} = \frac{1}{12}$, so it alone takes 12 hours.